

London School of Economics
EC400 Mathematics SOLUTIONS
September 2016

Time: 3 hours

This examination has three parts: Revision Mathematics, Mathematics for Microeconomics, and Mathematics for Macroeconomics. Each part is worth 100 marks. Write your name, your class group number, and the name of your class teacher on your script. Explain your answers to get high marks.

Part 1. Revision Mathematics. Answer all questions.

Some of these questions have yes no answers, or numerical answers. You must explain your answers to these questions.

Part 2. Mathematics for Microeconomics. Answer all questions.
Partial credit should be awarded, while accounting the quality of the answer!

5. (25 marks) Consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + bx_2^2 + 2abx_1x_2.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

Find all the principal submatrices of A .

Solutions: (2 Marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & ab \\ ab & b \end{pmatrix}$$

(2 Marks) Its principal submatrices are:

$$\begin{array}{ll} 2^{\text{nd}} \text{ Order:} & \begin{pmatrix} a & ab \\ ab & b \end{pmatrix} \\ 1^{\text{st}} \text{ Order:} & (a) \text{ and } (b). \end{array}$$

- (b) (8 marks) Define what it means for the quadratic form $Q(\mathbf{x})$ to be *positive semi-definite* and *negative semidefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(\mathbf{x})$ is positive semidefinite or negative semidefinite?

Solutions: (3 Marks) A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, and is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x}$.

(5 Marks) The quadratic form is positive semidefinite if all the principal minors are non-negative; and it is negative semidefinite if:

- all even principal minors of A are non-negative;
- all odd principal minors of A are non-positive.

- (c) (9 marks) For what values of the parameter values, a and b , is the quadratic form $Q(\mathbf{x})$ *indefinite*? Plot your answer in $\mathbb{R}^2$.

Solutions: (9 Marks) $Q(\mathbf{x})$ is indefinite if it is neither positive semidefinite nor negative semidefinite. $Q(\mathbf{x})$ is positive semidefinite if all principal minors are non-negative which requires:

$$a \geq 0, \quad b \geq 0, \quad ab(1 - ab) \geq 0.$$

For these to hold at once it must be that

$$a \geq 0, \quad b \geq 0, \quad 1 \geq ab.$$

$Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

$$a \leq 0, \quad b \leq 0, \quad ab(1 - ab) \geq 0.$$

For these to hold at once it must be that

$$a \leq 0, \quad b \leq 0, \quad 1 \geq ab.$$

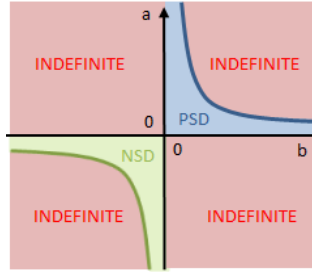
Thus the quadratic form is indefinite if and only if

$$ab > 1 \quad \text{or} \quad ab < 0.$$

(d) (4 marks) Plot your answer in $\mathbb{R}^2$.

Solutions: (4 Marks)

The plot below depicts the answer:



1. (25 marks) Answer the following questions:

(a) (7 marks) Provide two equivalent definitions of *quasi-concavity*.

Solutions: (1 Mark) Let f be a function defined on a convex set $U \subset \mathbb{R}^n$. (3 Marks) The function f is quasi-concave if upper-contour sets $C_k^+ = \{x | f(x) > k\}$ are convex for all values of k . (3 Marks) The function f is quasi-concave if for all $\mathbf{x}, \mathbf{y} \in U$ and $t \in [0, 1]$,

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

(b) (10 marks) Show that any continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$x \frac{\partial f(x)}{\partial x} \leq 0,$$

must be *quasi-concave*. Prove your answer.

Solutions: (10 Marks) First observe that the conditions in the statement implies that the function always increases in the negative orthant, while it decreases on the positive orthant as

$$\begin{aligned} x < 0 & \text{ implies } \frac{\partial f(x)}{\partial x} \geq 0; \\ x > 0 & \text{ implies } \frac{\partial f(x)}{\partial x} \leq 0. \end{aligned}$$

Thus, the function must have a maximum at $x = 0$. Next we show that implies that any non-empty upper-contour set must be an interval. Take a value $k \leq f(0)$. Look for the smallest value $x \leq 0$ such that

$$f(x) = k.$$

If it exists call it $\underline{x}_k$, otherwise set $\underline{x}_k = -\infty$. By definition of $\underline{x}_k$ and continuity, for any $x \leq 0$, we have that

$$f(x) \geq k \text{ if and only if } x \geq \underline{x}_k.$$

Next look for the largest value $x \geq 0$ such that

$$f(x) = k.$$

If it exists call it $\bar{x}_k$, otherwise set $\bar{x}_k = \infty$. By definition of $\bar{x}_k$ and continuity, for any $x \geq 0$, we have that

$$f(x) \geq k \text{ if and only if } x \leq \bar{x}_k.$$

Thus, $C_k^+ = [\underline{x}_k, \bar{x}_k]$ for all k , which is an interval and thus a convex set. So, the function is quasi-concave.

The function is not properly single-peaked as the condition does not hold strictly away from $x = 0$. Arguments without an explicit proof, or definition of single-peak should be awarded only partial credit, at most 6 Marks.

- (c) (8 marks) Find an *increasing* function $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ that is *not quasi-concave*. Prove your answer.

Solutions: (8 marks) Consider the function

$$f(x, y) = x^2 + y^2.$$

Such function is clearly increasing in $\mathbb{R}_+^2$ as $Df(x, y) = (2x, 2y)$. However, the function is not quasi-concave. In fact, $(1, 0), (0, 1) \in C_1^+$. However, $(1/2, 1/2) \notin C_1^+$ as

$$f(1/2, 1/2) = 1/2 < 1.$$

7. (50 marks) For $a > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & ax + y \\ \text{s.t.} \quad & x^2 + (x - y)^2 \leq 1 \\ & x \geq a, y \geq 0. \end{aligned}$$

- (a) (6 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither.

Solutions: (3 Marks) The objective function, as well as the last two constraints are obviously linear and thus both convex and concave. (3 Marks) The first constraint is strictly convex, as its Hessian is positive definite for $x, y \geq 0$,

$$D^2F(x, y) = \begin{pmatrix} 4 & -2 \\ -2 & 2 \end{pmatrix}, \quad |D^2F(x, y)| = 4 > 0 \text{ and } 4 > 0.$$

- (b) (6 marks) Set up the constrained optimization problem using the Lagrangean function. Can you solve this optimization problem using the Kuhn-Tucker method? That is, for what values of a is Slater condition satisfied?

Solutions: (2 Marks) The Lagrangean of the problem is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}|a) = ax + y - \lambda_1(x^2 + (x - y)^2 - 1) - \lambda_2(a - x) + \lambda_3y.$$

(4 Marks) When the constraint set has a non-empty interior, we can adopt the Kuhn-Tucker approach to solve the problem. This is the case when $a < 1$ as in that case setting $x = y = a + \varepsilon$ is in the interior of the constraint set for $\varepsilon < (1 - a)$. When $a \geq 1$, no point satisfies all the constraints strictly as $x > a$, implies

$$x^2 + (x - y)^2 \geq a^2 \geq 1,$$

which delivers the result.

For parts (c), (d), and (e) suppose that Slater condition is satisfied.

- (c) (8 marks) Using the Kuhn-Tucker approach, write down the necessary first order conditions for the solution of the constrained optimization problem. Are solutions to these conditions also maximizer of the Lagrangean?

Solutions: (6 Marks) The necessary first order conditions for the solution of the constrained maximization problem are:

$$\begin{aligned} 0 &= a - \lambda_1(4x - 2y) + \lambda_2, \\ 0 &= 1 - \lambda_1(2y - 2x) + \lambda_3, \\ 0 &= \lambda_1(x^2 + (x - y)^2 - 1), \\ 0 &= \lambda_2(x - a) = \lambda_3y, \\ 0 &\geq x^2 + (x - y)^2 - 1, \quad 0 \geq a - x, \quad 0 \geq -y, \\ 0 &\leq \lambda_1, \quad 0 \leq \lambda_2, \quad 0 \leq \lambda_3. \end{aligned}$$

(2 Marks) Any solutions to these conditions is maximizer of the Lagrangean as the Lagrangean is a sum of concave functions by the arguments in (a) and (b).

(d) (18 marks) Solve the constrained optimization problem in terms of a .

Solutions: Consider the various supports:

- (3 Marks) At an optimum it also cannot be that $\lambda_1 = 0$, as the FOC for good x would imply that, $\lambda_2 = -a < 0$. So there is no solution on these supports.
- (4 Marks) At an optimum it also cannot be that $\lambda_3 > 0$, as the FOC for good y would imply that

$$\lambda_3(a) = -(1 + 2\lambda_1(a)x(a)) \leq -1 < 0,$$

by $\lambda_1(a) \geq 0$ and $x(a) \geq 0$. So there can be no solution on these supports.

- (5 Marks) At an optimum it also cannot be that $\lambda_1, \lambda_2 > 0$ while $\lambda_3 = 0$. If so, the two constraints would imply that

$$x(a) = a \quad \text{and} \quad y(a) = a + (1 - a^2)^{1/2}.$$

For the solution to be well defined $a < 1$ which we imposed via Slater condition. If so, the FOC for good y would imply that

$$\lambda_1(a) = \frac{1}{2(1 - a^2)^{1/2}} > 0.$$

If so,

$$\begin{aligned} \lambda_2(a) &= \lambda_1(a)(4x - 2y) - a = \frac{a}{(1 - a^2)^{1/2}} - 1 - a \geq 0 \\ \Leftrightarrow \quad a^4 + 2a^3 + a^2 &\geq 2a + 1, \end{aligned}$$

which is equivalent to $a \geq 0.8832$. Thus, when a is sufficiently high this must be the solution.

- (6 Marks) Thus, in the remaining case the optimum must live in the only remaining support, namely $\lambda_1 > 0$ and $\lambda_3 = \lambda_2 = 0$. Solving FOC at such values one finds that:

$$x(a) = \frac{1 + a}{((1 + a)^2 + 1)^{1/2}}, \quad y(a) = \frac{(2 + a)}{(1 + a)}x(a) > 0, \quad \lambda_1(a) = \frac{1 + a}{2x(a)} > 0.$$

For this to be a solution it must be that

$$x(a) \geq a \quad \Leftrightarrow \quad a^4 + 2a^3 + a^2 \leq 2a + 1.$$

Thus, as this is the only point satisfying all the necessary conditions, it is the unique solution for all $a \leq 0.8832$.

(e) (12 marks) State the Envelope Theorem in the context of this problem. Then, find the slope of value function with respect to a without relying on the Envelope Theorem.

Solutions: (6 Marks) The Envelope Theorem in general states that if $(\mathbf{x}(a), \boldsymbol{\lambda}(a))$ are differentiable functions and that $\mathbf{x}(a)$ satisfies the constraint qualification. If so,

$$\frac{dv(a)}{da} = \frac{\partial L(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = x(a) - \lambda_2(a) = x(a) = \sqrt{\frac{1+a}{a}}.$$

(6 Marks) To derive the result without relying on the theorem, denoting by f the objective function and by g the first constraint, observe that

$$\begin{aligned} \frac{dv(a)}{da} &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] + \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = \\ &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] + \lambda_1(a) \left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = \\ &= \left[\frac{\partial f(\mathbf{x}(a)|a)}{\partial a} \right] - \lambda_1(a) \frac{\partial g(\mathbf{x}(a)|a)}{\partial a} = \frac{\partial \mathcal{L}(\mathbf{x}(a), \boldsymbol{\lambda}(a)|a)}{\partial a} = \\ &= x(a), \end{aligned}$$

where the first equality is definitional, the second exploits FOC

$$\frac{\partial f(\mathbf{x}(a)|a)}{\partial \mathbf{x}} = \lambda_1(a) \left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right],$$

while the third exploits the binding constraint by totally differentiating it

$$\left[\frac{\partial g(\mathbf{x}(a)|a)}{\partial \mathbf{x}} \right]^T \frac{\partial \mathbf{x}(a)}{\partial a} = -\frac{\partial g(\mathbf{x}(a)|a)}{\partial a}.$$

Thus, the value function increases as a increases, for all $a \leq 0.8832$. If instead $a \in (0.8832, 1)$ and $\lambda_2(a) > 0$, a similar logic establishes that

$$\frac{dv(a)}{da} = x(a) - \lambda_2(a) = 2a + 1 - \frac{a}{(1-a^2)^{1/2}}.$$

More direct answers to this part are possible and **should be awarded full credit**.

Part 3. Mathematics for Macroeconomics. Answer all questions.

8. **(50 marks)**